FCFS

Q1.

Consider the following set of processes having their burst time mentioned in milliseconds. CPU burst time indicates that for how much time, the process needs the cpu.

Process	Burst Time
P1	24
P2	3
P3	3

Calculate the average waiting time if the processes arrive in the order of:
a). P1, P2, P3

a. The processes arrive the order P1, P2, P3. Let us assume they arrive in the same time at 0 ms in the system. We get the following gantt chart.

Waiting time for P1= 0ms , for P2 = 24 ms for P3 = 27ms
Avg waiting time: (0+24+27)/3= 17

Q2.

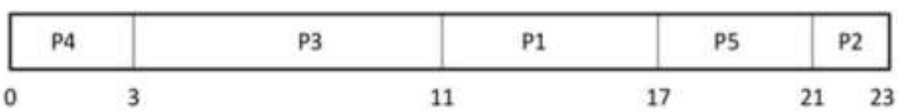
Process	Burst Time
P1	24
P2	3
P3	3

b.) If the process arrive in the order P2,P3, P1

Average waiting time: (0+3+6)/3=3. Average waiting time vary substantially if the process CPU burst time vary greatly.

Q3.

Process	Burst time	Arrival time
P1	6	2
P2	3	5
P3	8	1
P4	3	0
P5	4	4



P4 = 0-0 = 0

P3 = 3-1 = 2

PI = 11-2 = 9

P5= 17-4 = 13

P2= 21-5= 16

Average Waiting Time

$$\frac{0+2+9+13+16}{5}$$

= 40/5= 8